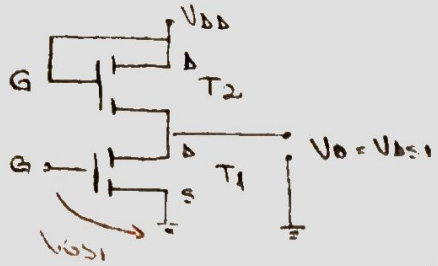
(B) Invertor NMOS cu R neliniar



$V_{GS2} = V_{DS2}$

T_1 și T_2 sunt identice

$K = K_1 = K_2 \quad V_{T1} = V_{T2} = V_T$

$i_{DS1} = i_{DS2}$ (toți param. tranzistor sunt identici)

$K = 20\mu A/V^2 \; ; V_{DD} = 6^V$

Ridicăm curbe de sarcină

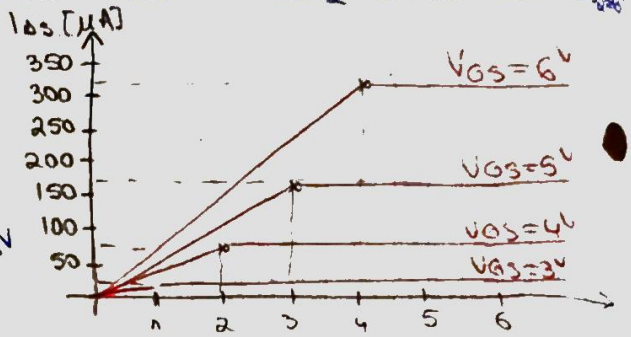
a) $V_{GS} = V_i \leq 2^V = V_T$ avem $i_{DS1} = 0 \Rightarrow i_{DS2} = 0 \Rightarrow V_{DS2} = 0 \Rightarrow V_O = V_{DD} - V_{DS2} = 6^V$

b) $V_{GS1} = 3^V \Rightarrow i_{DS1} = i_{DS2} = 20\mu A$

Pt. $i_{DS2} = 20\mu A \Rightarrow V_{GS2} = 3^V$

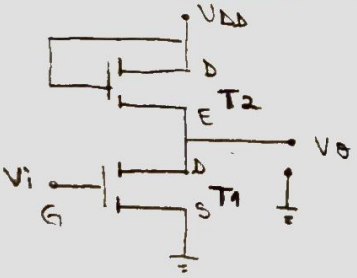
Dar $V_{GS2} = V_{DS2} = 3^V$

$\Rightarrow V_O = V_{DS1} = V_{DD} - V_{DS2} = 6 - 3 = 3^V$



CURS 3

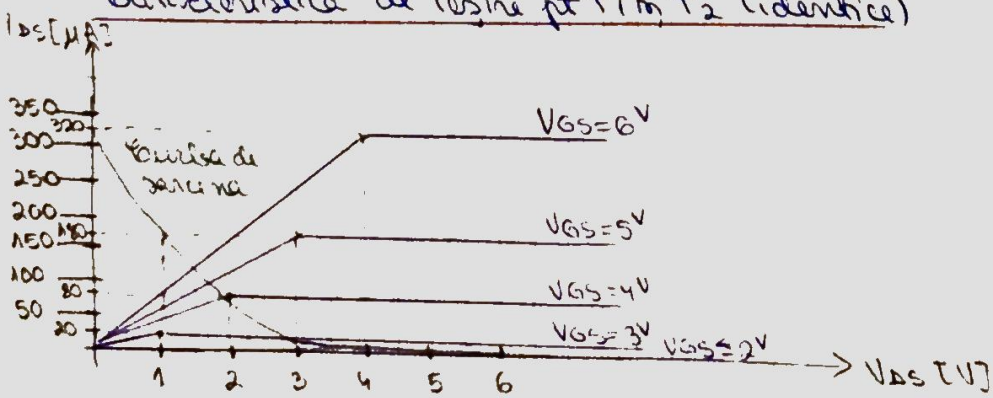
(B) Invertor NMOS cu R neliniar (continuare)



T_2 e echivalent cu T_1 dacă $\begin{cases} K_1 = K_2 = K \\ \frac{W_1}{L_1} = \frac{W_2}{L_2} = 1 \\ V_{T1} = V_{T2} = V_T \end{cases}$

$i_{DS} = 2K\left[(V_{GS} - V_T)V_{DS} - \frac{V_{DS}^2}{2}\right]$ pt reg liniar

$i_{DS} = K[V_{GS} - V_T]^2$ pt. reg. saturat $\rightarrow i_{DS}$ cel mai mare

$K = 20\mu A/V^2 \; ; V_{DD} = 6V \; ; V_T = 2V$

Caracteristica de ieșire pt T_1 și T_2 (identice)